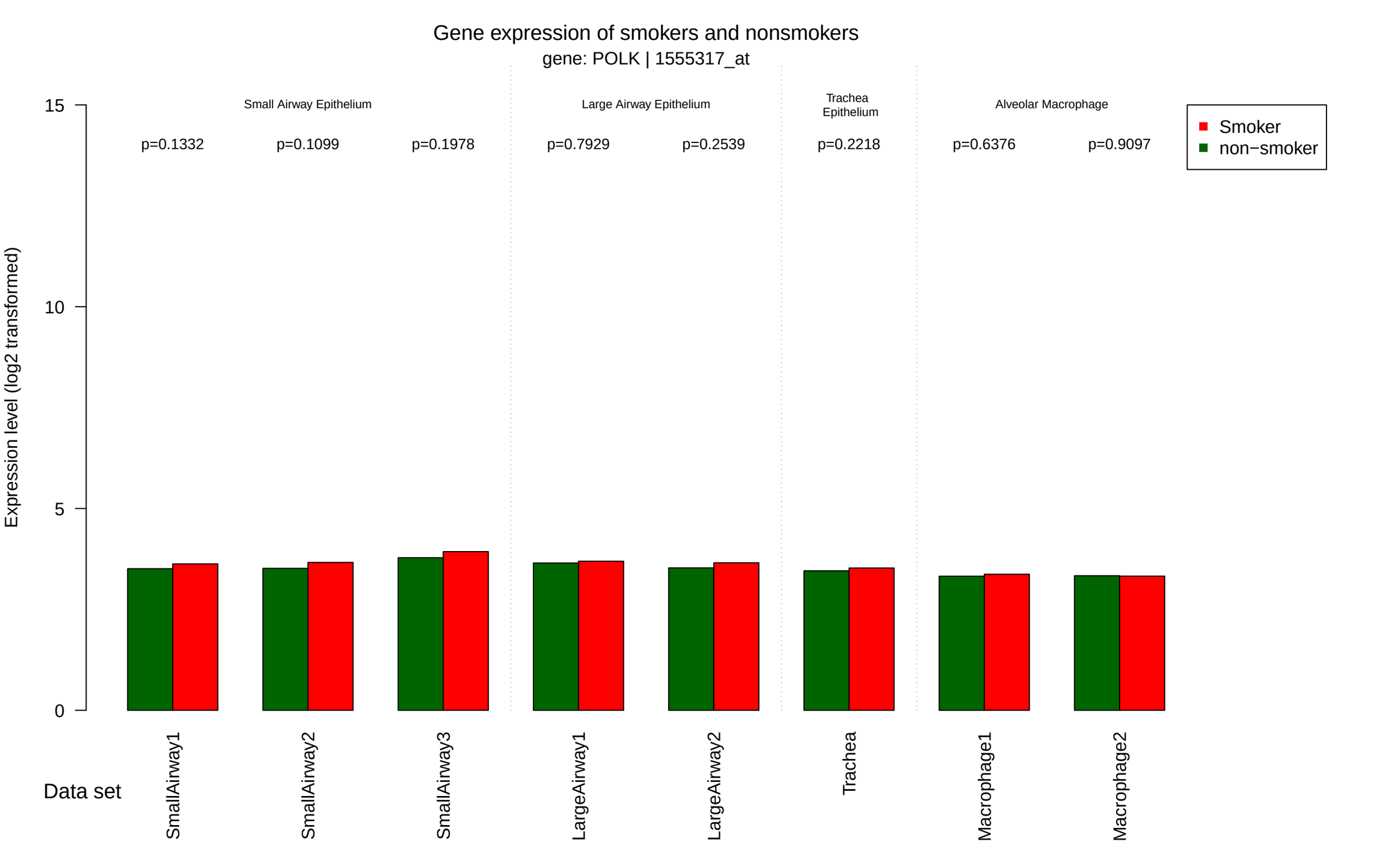
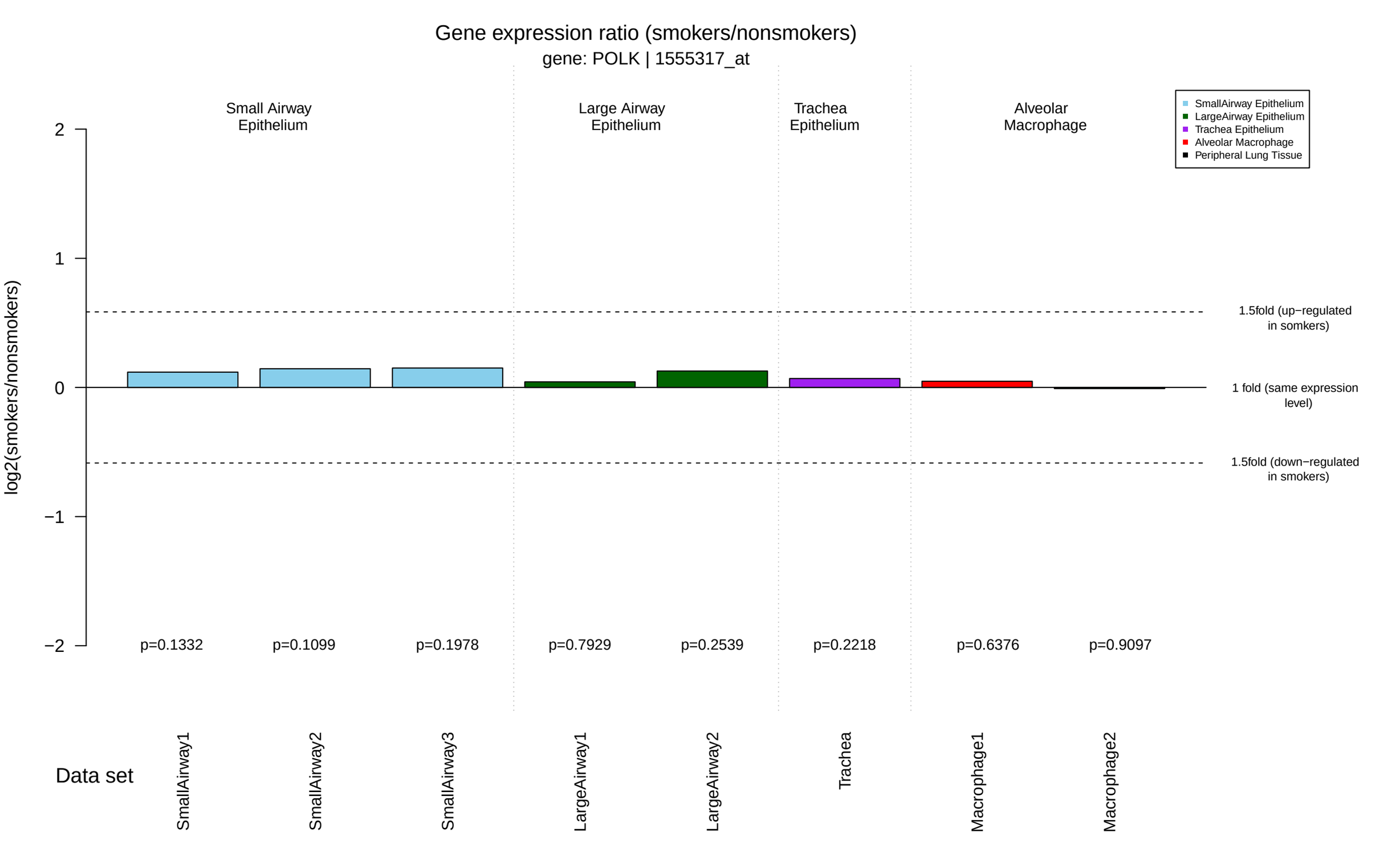
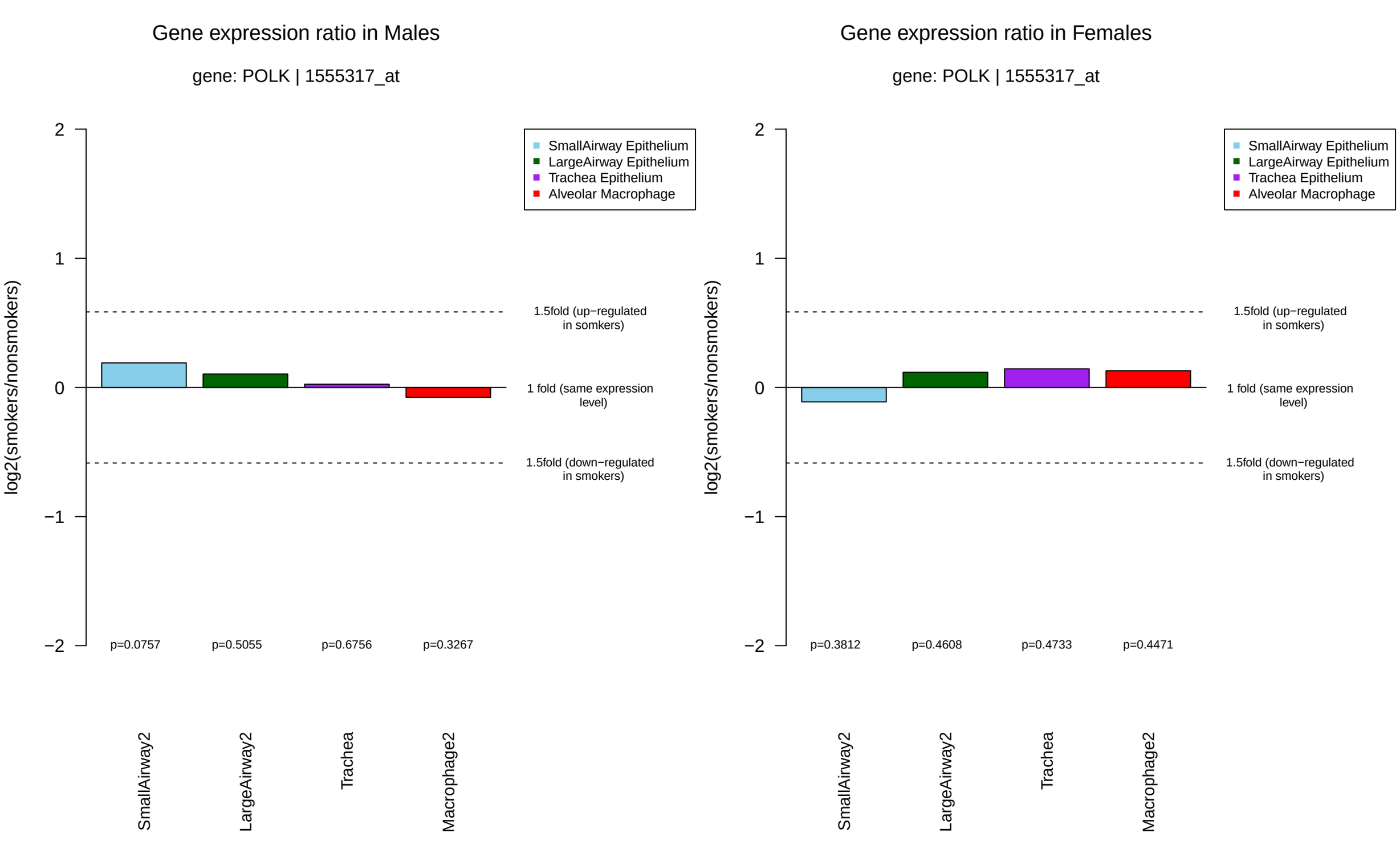
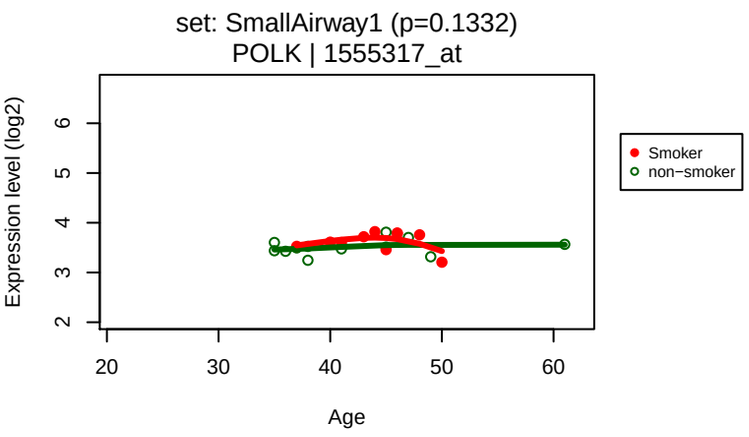
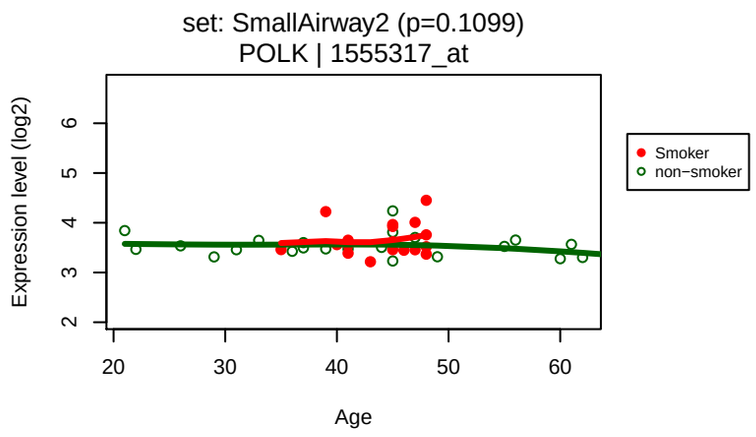
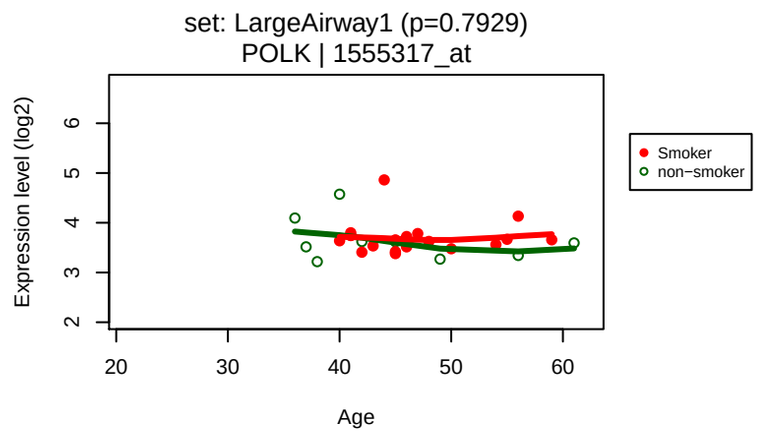

Gene expression of smokers and nonsmokers
gene: POLK | 1555317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1332
p=0.1099
p=0.1978
p=0.7929
p=0.2539
p=0.2218
p=0.6376
p=0.9097
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

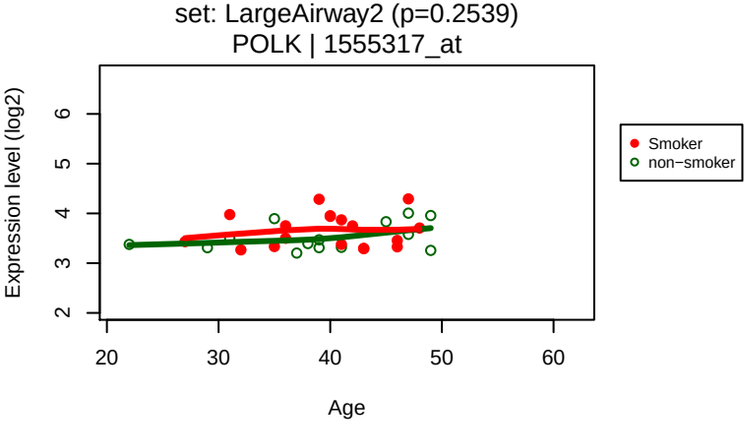

Gene expression ratio (smokers/nonsmokers)
gene: POLK | 1555317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1332
p=0.1099
p=0.1978
p=0.7929
p=0.2539
p=0.2218
p=0.6376
p=0.9097
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

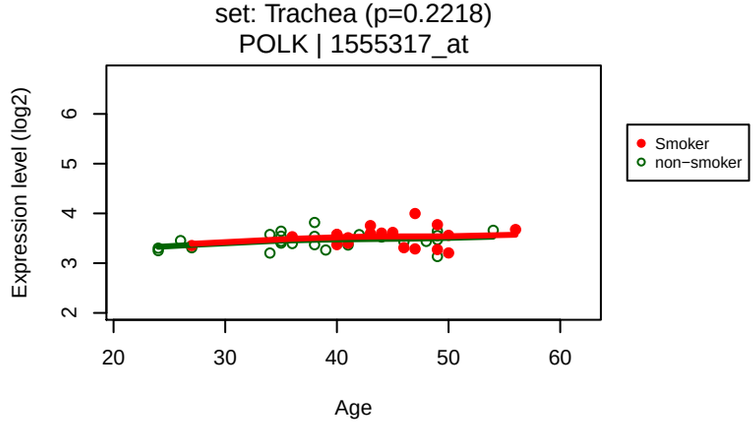

Gene expression ratio in Males
gene: POLK | 1555317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0757
p=0.5055
p=0.6756
p=0.3267
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POLK | 1555317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3812
p=0.4608
p=0.4733
p=0.4471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

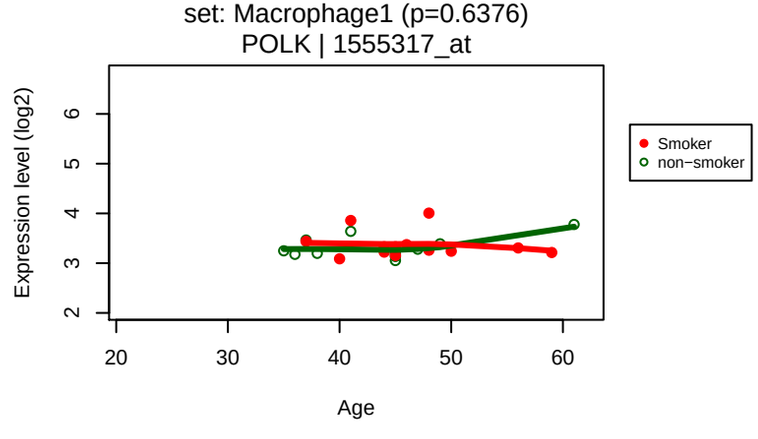

set: SmallAirway1 (p=0.1332)
POLK | 1555317_at
Expression level (log2)
Age
Smoker
non−smoker

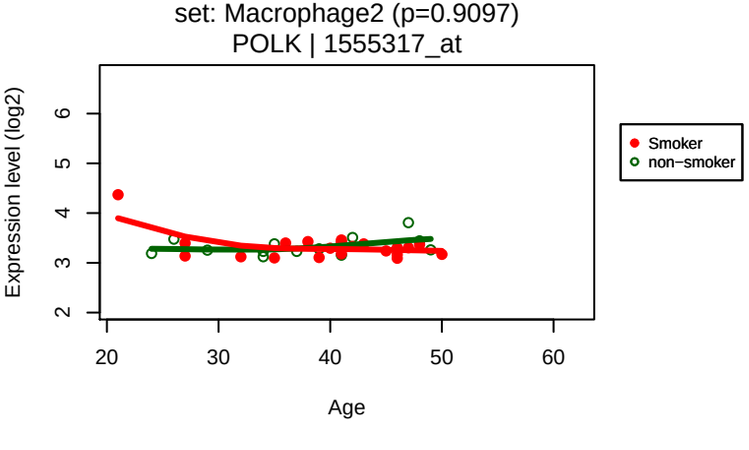

set: SmallAirway2 (p=0.1099)
POLK | 1555317_at
Expression level (log2)
Age
Smoker
non−smoker

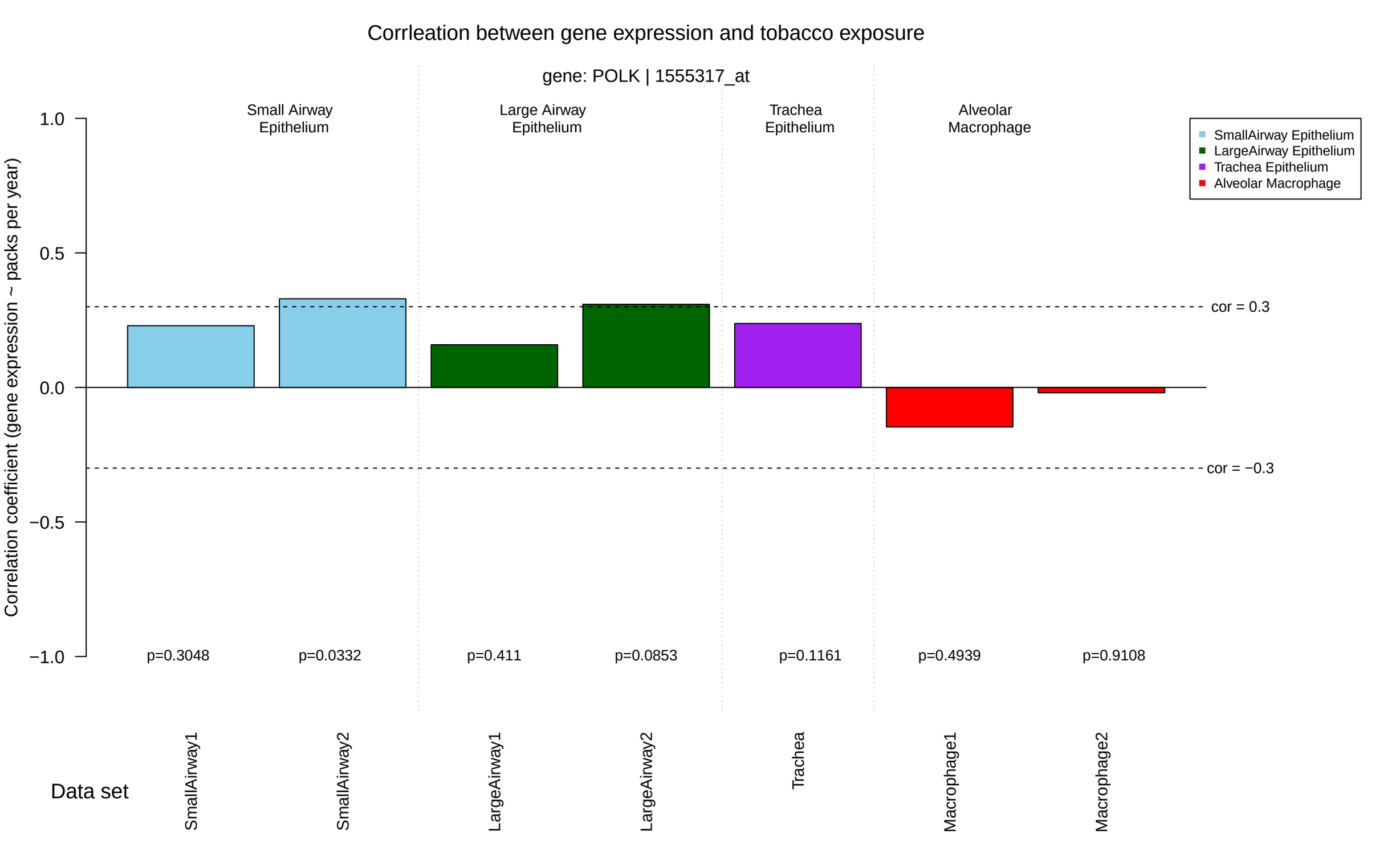

set: LargeAirway1 (p=0.7929)
POLK | 1555317_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2539)
POLK | 1555317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2218)
POLK | 1555317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6376)
POLK | 1555317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9097)
POLK | 1555317_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POLK | 1555317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3048
p=0.0332
p=0.411
p=0.0853
p=0.1161
p=0.4939
p=0.9108
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2